Quiz 4 Answers

1a) $S \to SS$ will give L^+. L^* must have ε so add the rule $S \to \varepsilon$ since we don't know whether $\varepsilon \in L$.

1b) To show suffix(L) is a CFL we can create a PDA or a CFG for suffix(L).

PDA soln

For $w \in L$ let $w = uv$. Then an accepting path for w in the PDA accepting L will look like

$$\longrightarrow q_0 \overset{u}{\dashrightarrow} q \overset{v}{\dashrightarrow} q_f$$ where q_f is a final state

if we reach state q with v as input and an identical stack then v will be accepted.

We do this as follows:

1. Make a copy M' of PDA M accepting L.
2. Replace every transition that consumes i/p with one that ignores it:

$$\sigma, \gamma \to \gamma' \implies \varepsilon, \gamma \to \gamma'$$

3. From each state in M' create transition $\varepsilon, \varepsilon \to \varepsilon$ to corresponding state in M.
4. Let q_0' (corresp. to q_0 in M) be the start state.
5. The final states are those in M.

The combination M, M' is clearly a PDA that accepts suffix(L).

<u>Grammar soln</u>

Let CFG G generate L. Assume G is in CNF.
We construct a CFG grammar G' that will generate all suffixes for $w \in L$.

For every variable X in G create a corresponding var X'. For every rule $X \to YZ$ add the following rules to G':

$$X' \to Y'Z, \quad X' \to Z, \quad X' \to X, \quad X' \to \varepsilon$$

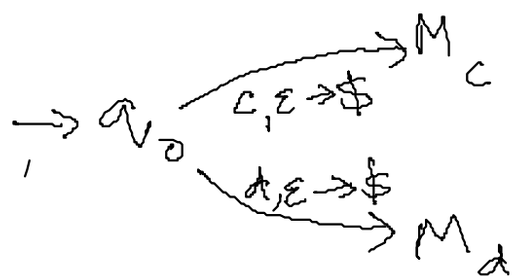
It is clear X' generates suffixes of all strings derived by X.

$G' = (V \cup \{X' \mid X \in V\}, \Sigma, P \cup \text{rules mentioned above}, S')$

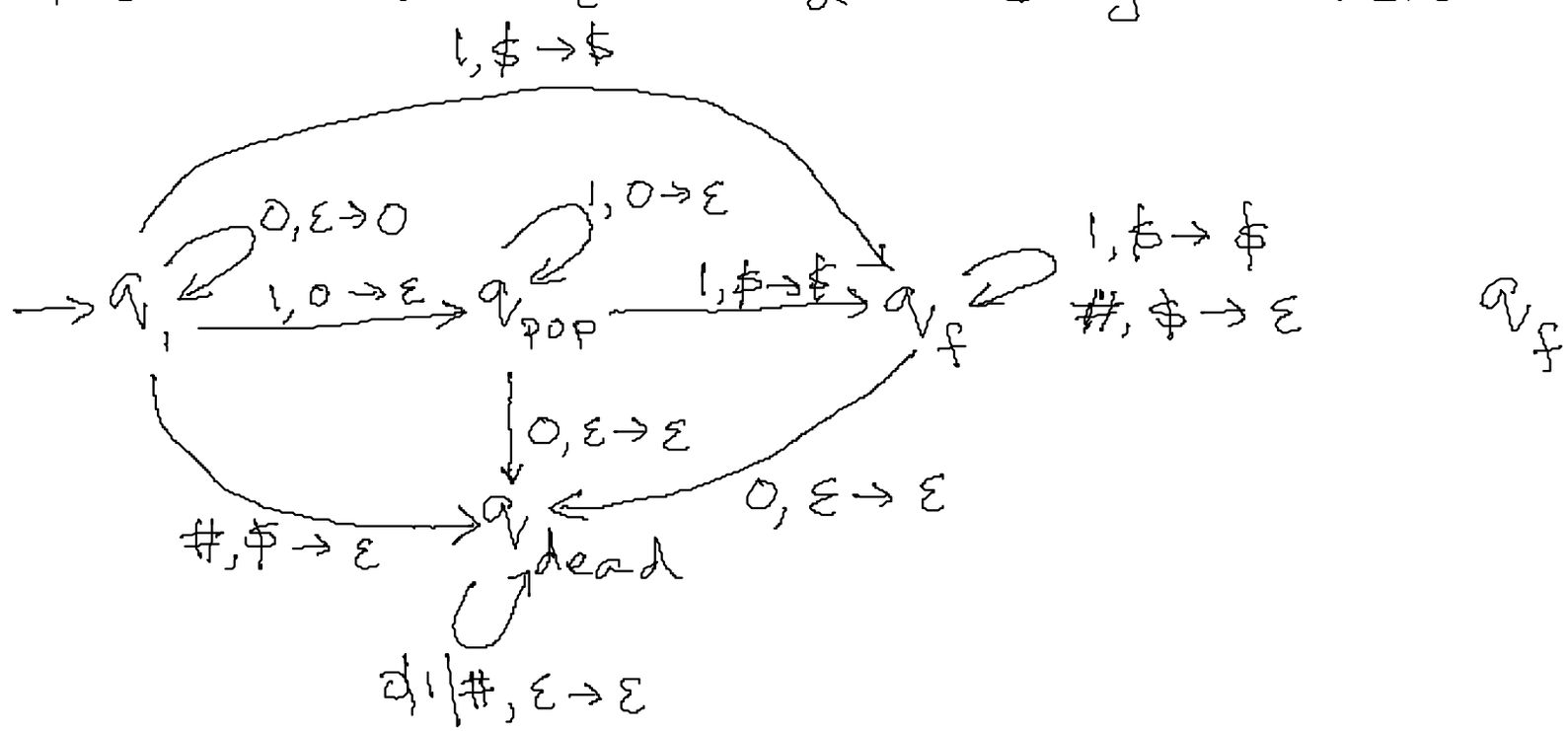
where $G = (V, \Sigma, P, S)$.

Q2 Let M_c, M_d be DPDAs for the two sub-languages where strings start with c, d resp.

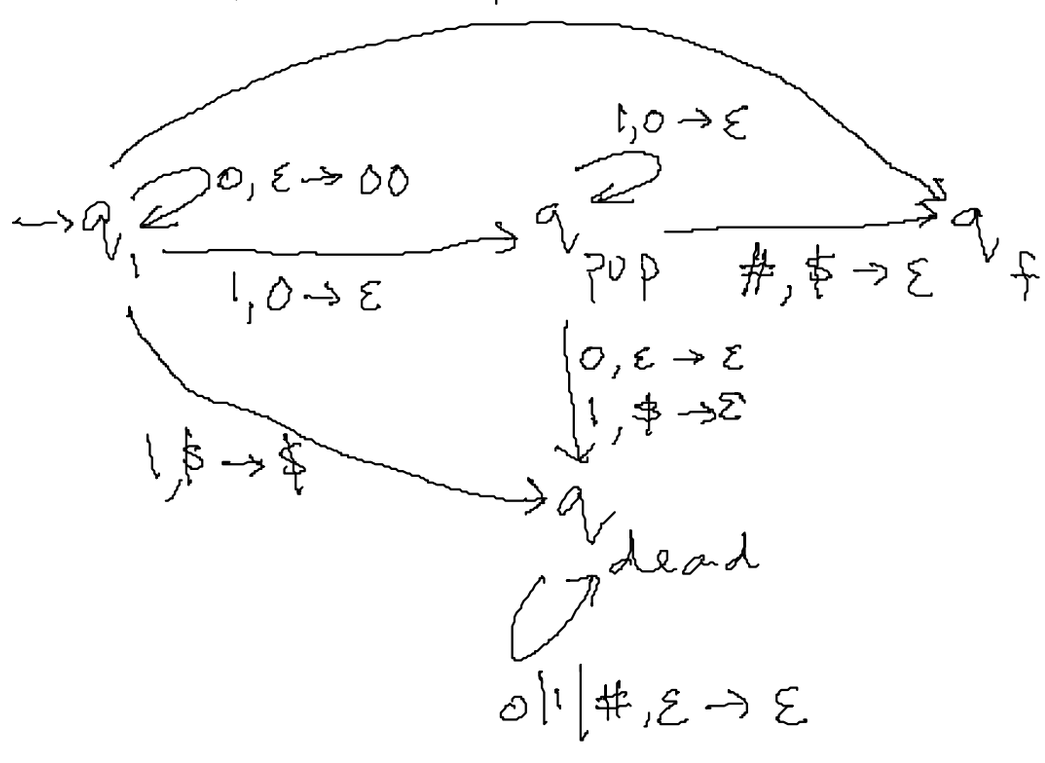
Then the DPDA for $L(M_c) \cup L(M_d)$ is



The DPDAs M_c & M_d are given below. (# added at end)



M_d



From the above L is a DCFL.
if c, d are not present then we have non-determinism from q_0 to M_c & M_d and the resulting lang. is not DCF but just CF.
This also shows that DCFLs are not closed under union.

Q3 a) Informal: L is the language of legal brackets
– replace 0 by '(' and 1 by ')'.

Formal: $L = \{w \in \{0,1\}^* \mid$ No. of 0s in every prefix of $w \geq$ No. of 1s$\}$

Most of you have not got both parts correct.
Partial credit given.

Q3b) The easy and lazy answer is:
since ε has multiple parses G is ambiguous and so not a DCFG.
However, for all $w \in L$ we can always get a unique leftmost redn by sticking to replacing ε by S instead of SS, and single lookahead.

For example: consider 0011

$0011 \to 00S11 \to 0S1 \to S$

Another more complex example: 00101101

$00101101 \to 00S101101 \to 0S01101 \to 0S0S1101$

$\to 0SS101 \to 0S101 \to S01 \to S0S1 \to SS \to S$

The ambiguity is a spurious ambiguity because the correct parse tree is a subtree of the other ambiguous parses.

Also, it is fairly easy to construct a DPDA for L so it is a DCFL.

Lazy answer gets no credit.
if DPDA drawn – partial credit.
For full credit have to argue that for any w one rule, same rule applies for a leftmost redn.

Q4a) Let $L = L(G)$ where G is a CFG in CNF.

Then for the homomorphism $h : \Sigma \to \Sigma_h^*$ we construct a CF grammar G_H that generates $h(L)$.

For all rules of the form $X \to \sigma$, $\sigma \in \Sigma$ in G replace it by $X \to h(\sigma)$. So $G_H = (V, \Sigma_h, P_h, S)$ where P_h contains all rules in P with the above changes. It is clear that $L(G_h) = h(L)$.

Q4b) For this question most of you have copied the answer from the same original source. There are many possible answers. The one that

is immediately obvious is the example language in Q2. With $h(c) = h(d) = \varepsilon$ the language $h(L)$ is not a DCFL but a CFL.

Many of you have used:

$$L_1 = \{a^m b^m c^n \mid m, n \geq 0\}$$

$$L_2 = \{a^m b^n c^n \mid m, n \geq 0\}$$

$$L = dL_1 \cup L_2, \quad h(d) = \varepsilon, \quad h(\sigma) = \sigma.$$

and argued that $h(L)$ is not a DCFL. For this you must prove that L is a DCFL. No one has done that. DCFLs are not closed under union and concatenation.

Also L is not $\{a^m b^m c^m \mid m \geq 0\}$ rather it is $\{a^m b^n c^p \mid \text{either } m = n \text{ or } n = p\}$ which CF.

No credit if you have not proved L is a DCFL.